

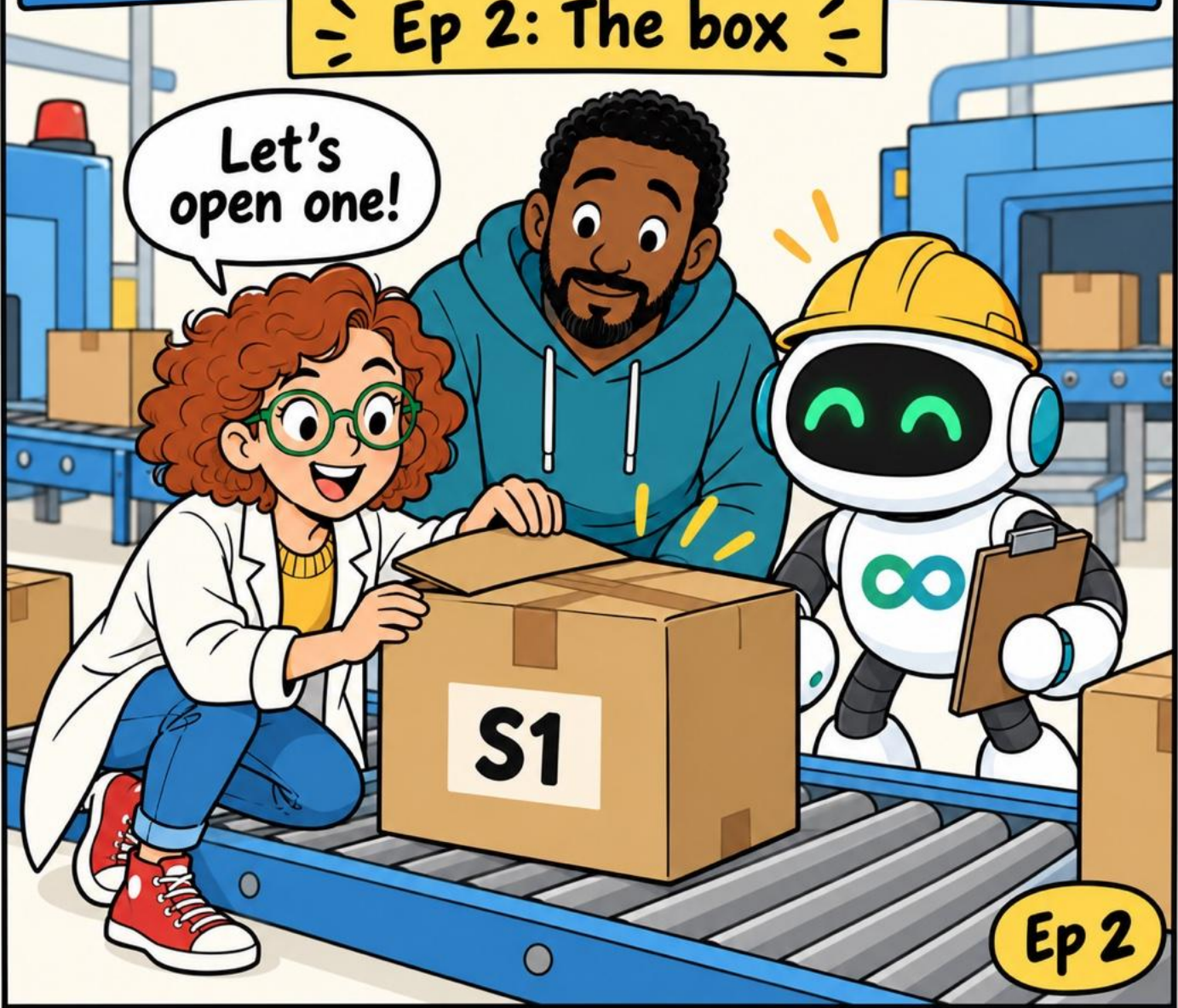
THE PIPELINE FACTORY

Ep 2: The box

Let's
open one!

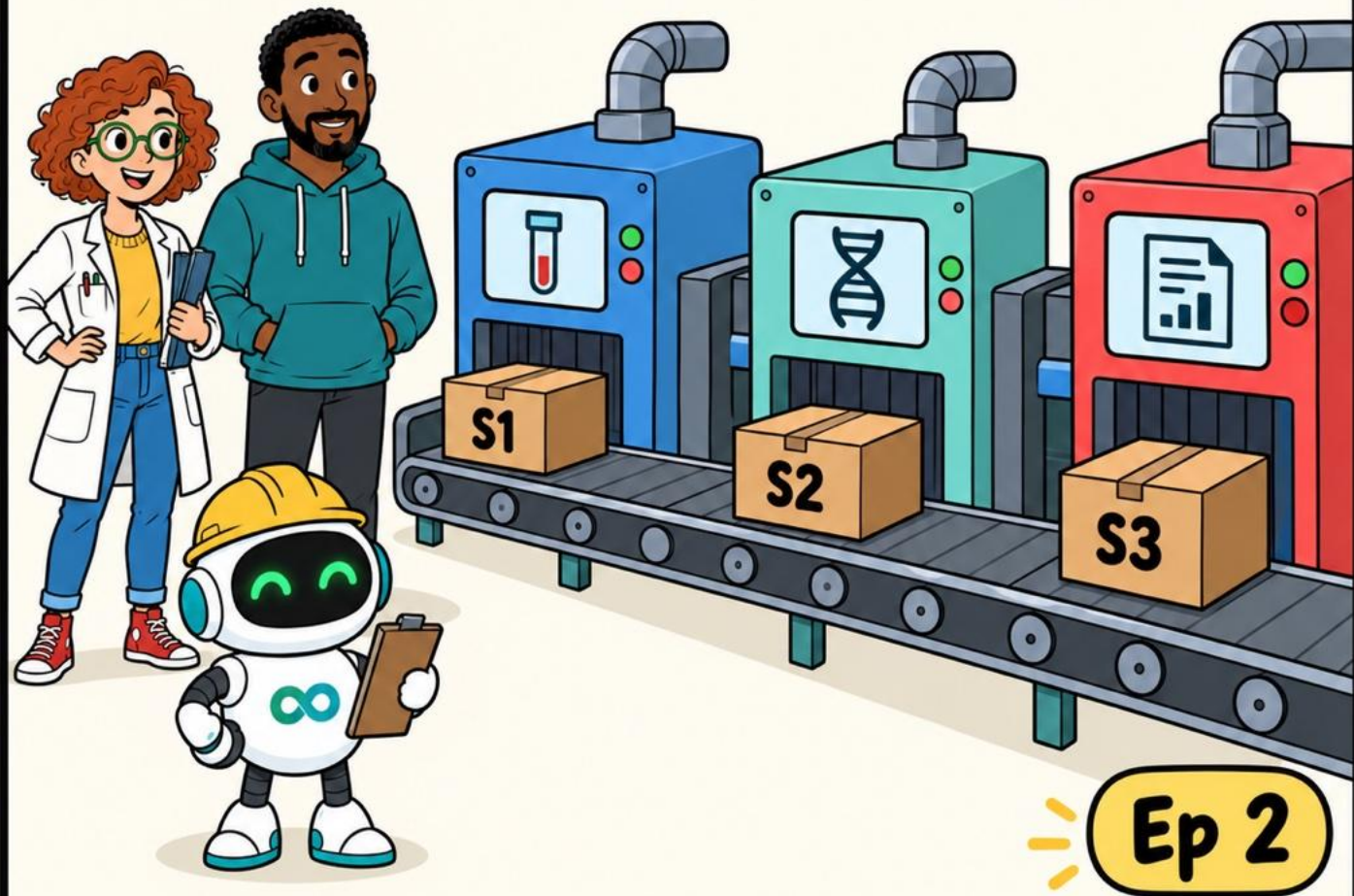
S1

Ep 2



≡ LAST TIME ≡

A factory: stations do the jobs, belts carry boxes of samples between them.



THE LABEL

What's this box?

Sample S1.
Read the label.

```
val(sample_id) = S1
```

val = a plain
value, like text.
Here: the name.

Ep 2

THE FILES

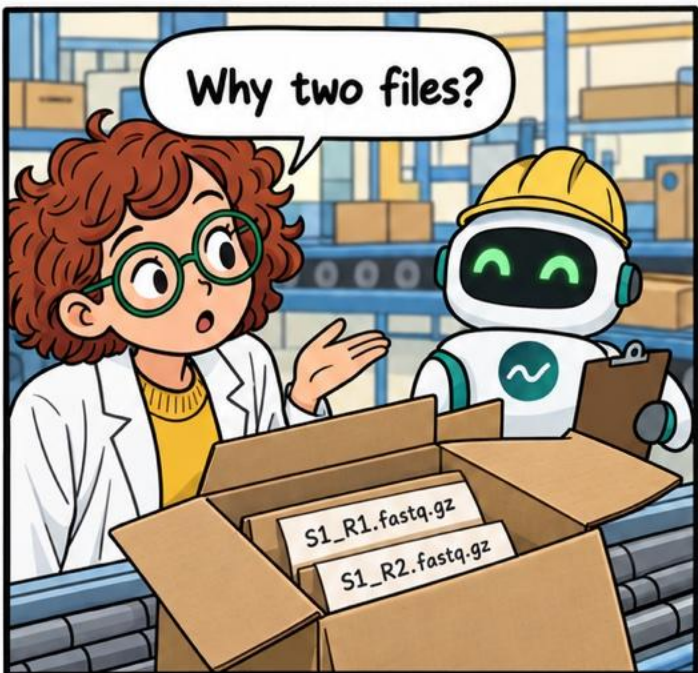
And path(reads)?



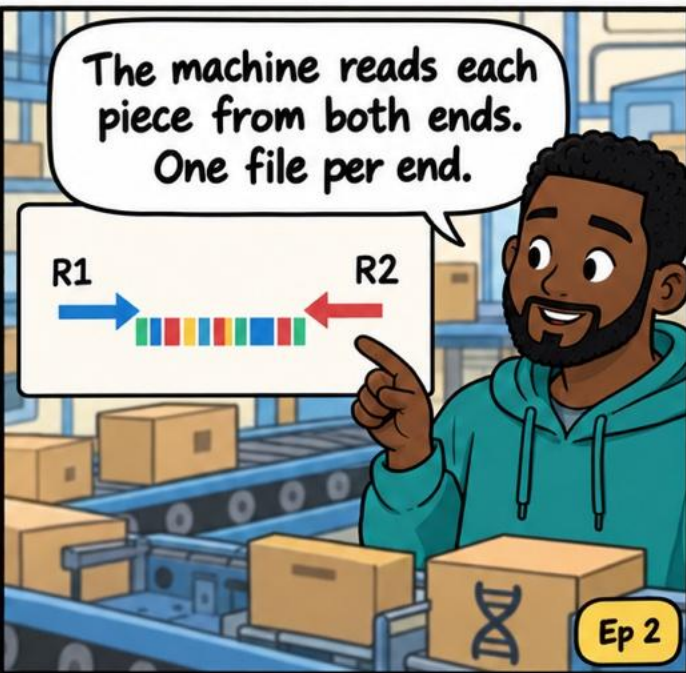
path = real files.
Nextflow brings them
to the station.



Why two files?



The machine reads each
piece from both ends.
One file per end.



THE TAPE



```
["S1", [S1_R1.fastq.gz, S1_R2.fastq.gz]]
```

≡ WHO WROTE "S1"? ≡

Who wrote 'S1'
on the box?

Nextflow, from
the file names:

```
data/*_R{1,2}.fastq.gz
```

Whatever * matches
is the name.

liver_R1.fastq.gz → liver

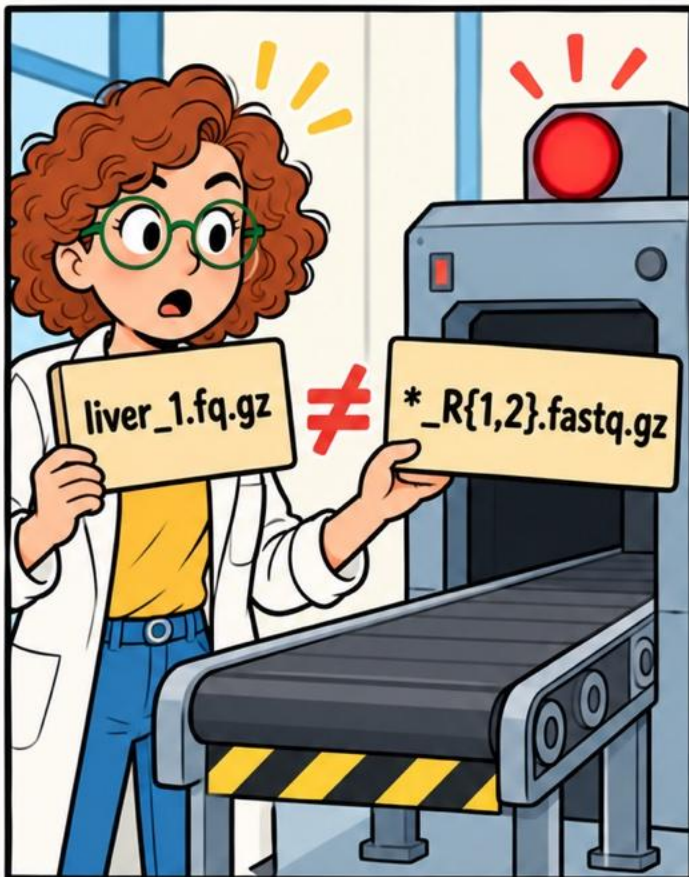
liver_R1.fastq.gz

S1

liver

Ep 2

WHEN IT GOES WRONG



**Pattern matches no file?
Nextflow stops. Fix it in
the control room.**



**Odd name or no partner file?
Skipped silently. Count your
boxes!**

THE BOX, IN CODE

```
1 // in nextflow.config (the control room):
2 params.reads = "data/*_R{1,2}.fastq.gz"
3
4 // in main.nf: this belt carries one box per sample:
5 // its name and its two read files
6 reads_ch = channel.fromFilePairs(params.reads,
7                                   checkIfExists: true)
8
9 // QUANT station, its input:
10 tuple val(sample_id), path(reads)
11 //     name: "S1"       files: R1 + R2
```



Ep 2

S1 first,
then S2?

No. Each box runs
at the same time, if
your computer has room.

QUANT

S1

QUANT

S2

QUANT

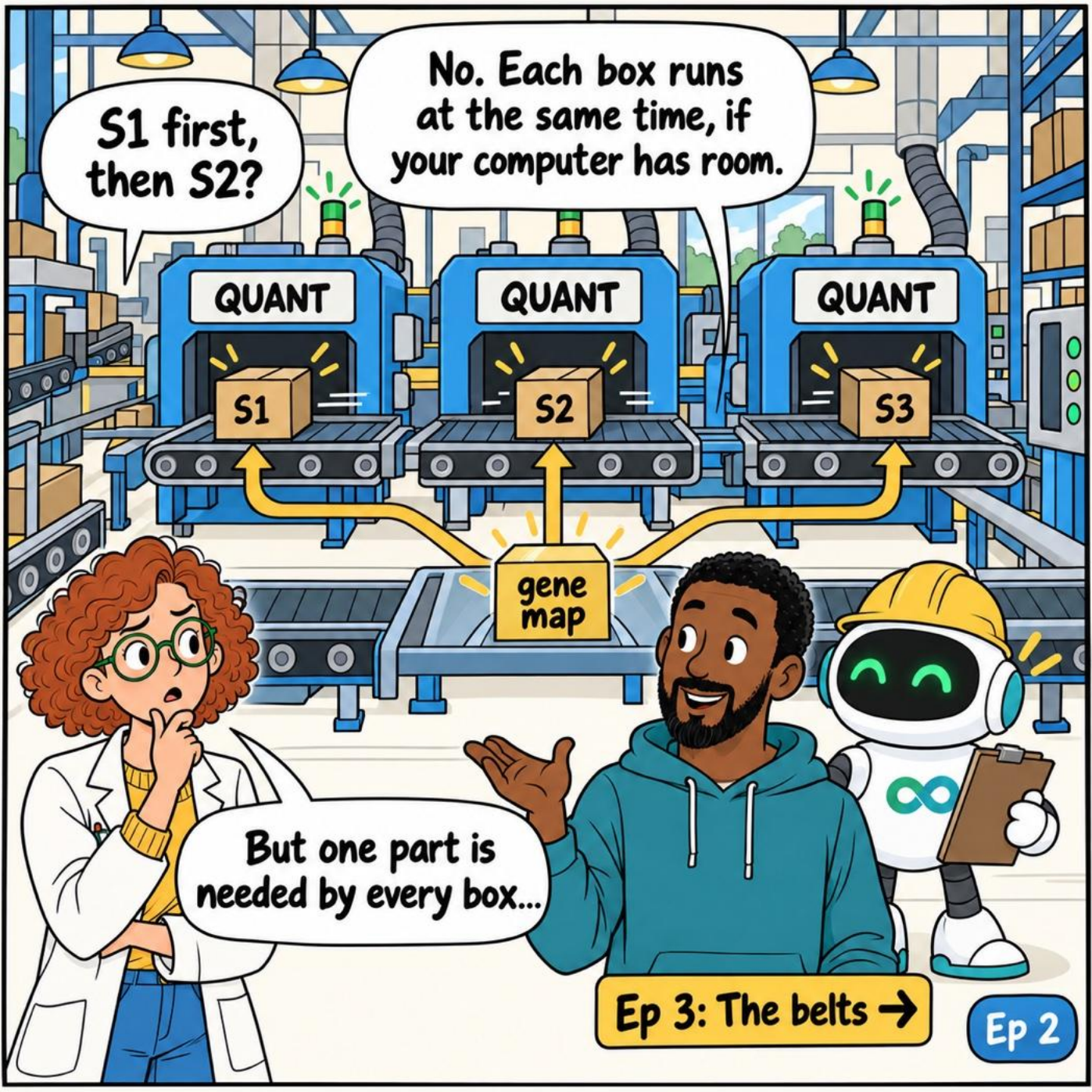
S3

gene
map

But one part is
needed by every box...

Ep 3: The belts →

Ep 2





THANKS FOR READING!

Follow me on GitHub,
where I explain how it all works:

<https://github.com/loukesio>